Problem 1

(a) False

(b) True

(c) False

(d) True

(e) False

Problem 2

(a) False. The investor is risk averse. Even though the Bernoulli utility is locally linear, but it is globally concave.

(b) False. The reward has to be at least \$1.

(c) True. The second lottery is riskier than the first one.

(d) False. The investor prefers L1 to L2.

(e) True.

1. Forget to judge whether the statement is true all fall. Will lose some points, depending on whether your explanation is valid. An answer is favorable If your explanation involves key words like linear and nonlinear, you will get full marks. Attention, we are talking about assets with uncertainty, and the utility involves expectation. Some students didn't consider this.
2. Just definition. In economics and finance, risk neutral preferences are preferences that are neither risk averse nor risk seeking. A risk neutral party's decisions are not affected by the degree of uncertainty in a set of outcomes, so a risk neutral party is indifferent between choices with equal expected payoffs even if one choice is riskier. Just a special case when the odd is fair.
3. Most students are right if they are familiar with the basic definition
4. X would like to forgive more for certainty. This tradeoff reveals that he hates risk.
5. SSD then FOSD, SSD is stronger. A counter example: C FOSD B, but C not SSD B.

State (die result)	1	2	3	4	5	6
Gamble A wins $	1	1	2	2	2	2
Gamble B wins $	1	1	1	2	2	2
Gamble C wins $	3	3	3	1	1	1

1. Some students forget to answer right and wrong. Lose some points. Not risk neutral, only locally. But globally concave, risk aversion.
2. The definition of reward confuse most of students. And the comparison is not attending L1 and attending L2 but attending L1 or not. Just (-2.3,3.7,0.5), and the expected utility is trivial. Don't worry since most students have problems in understanding the expression. You will get full marks if you computed the expected utility of participating in L1 correctly. 9.775.
3. You only need to answer this question intuitively, don't need rigorous mathematic proof. I notice some students discuss the range of rewards carefully, that is great, but you will get full marks as long as you point out the key point that the second lottery is risker.
4. Just calculate it. Not difficult right? Some students misunderstood that the threshold of the utility function is the initial wealth. I can't give you points.
5. Just as 4.

Set the equation and solve it. EU(Y0+X)=U(Y0+CE(X))

Problem 3

(1) With $\gamma = 1/2$,

$$\frac{[10 + \mathrm{CE}(\tilde{d})]^{1/2} - 1}{1/2} = \frac{1}{2}\frac{13^{1/2} - 1}{1/2} + \frac{1}{2}\frac{10^{1/2} - 1}{1/2},$$

thus,

$$\mathrm{CE}(\tilde{d}) = \left[\frac{1}{2}13^{1/2} + \frac{1}{2}10^{1/2}\right]^2 - 10 = 1.45.$$

(2) Similarly, with $\gamma = 2$

$$\mathrm{CE}(\tilde{d}) = \left[\frac{1}{2}13^{-1} + \frac{1}{2}10^{-1}\right]^2 - 10 = 1.30.$$

(3) Since the expected payoff from the risky asset is 1.5, the risk premium when $\gamma = 1/2$ is 0.05 and the risk premium when $\gamma = 2$ is 0.2.

Problem 4

(1)

With $Y = 100000$, the bet of 2000 represents two percent of the investor's initial income. Hence, with $\pi^* = 0.55$ and $k = 0.02$, the approximation

$$\pi^* \approx \frac{1}{2} + \frac{1}{4}kR_R(Y)$$

implies more specifically that

$$0.55 \approx \frac{1}{2} + (0.005)R_R(Y).$$

Hence, the investor's coefficient of relative risk aversion $R_R(Y)$ must be

$$R_R(Y) = \frac{0.55 - 0.50}{0.005} = 10.$$

(2)

The Bernoulli utility function

$$u(Y) = \frac{Y^{1-\gamma} - 1}{1 - \gamma}$$

implies a coefficient of relative risk aversion that is constant (independent of income) and given by $R_R(Y) = \gamma$. Hence, the answer to part (a), above, implies that $\gamma = 10$.

(3)

With $Y = 100000$, the new bet of 4000 represents four percent of the investor's initial income. Hence, with $k = 0.04$ and $R_R(Y) = 10$, the approximation

$$\pi^* \approx \frac{1}{2} + \frac{1}{4} k R_R(Y)$$

implies more specifically that

$$\pi^* \approx \frac{1}{2} + \frac{1}{4}(0.04)(10) = 0.50 + 0.10 = 0.60$$

is the probability of winning that makes the investor indifferent between accepting and rejecting the bet. Since the probability of winning that is actually offered is $\pi = 0.55$,